

# Benjamin Hardin

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**Interests:** *Human-Computer Interaction, Explainable AI, Physiological Sensing & LLMs*

**Skills:** C/C++, Python, Swift, SwiftUI, Java, Javascript, PyTorch, MySQL, Dart, React, ARM/x86-64 Assembly, OpenGL, Unity, Matlab, R, Agile/Scrum, Xcode, Git, Flutter, Unix

## Education

**University of Oxford**, Oxford, UK

*October 2022 – May 2027*

**Ph.D. in Computer Science**

*Advisors: Prof. Marina Jirotko & Prof. Lars Kunze*

*Thesis Topic: Psychological Safety and Trustworthiness of Autonomous Vehicles*

*Affiliated with: Oxford Robotics Institute, Responsible Technology Institute*

*Industry advisors: Capgemini Engineering, TRL (Transport Research Laboratory)*

**Purdue University**, West Lafayette, IN, USA

*August 2018 – May 2022*

**B.S. in Computer Science Honors**

**Mathematics Minor**

GPA: 3.81/4.00

**Certificate in Entrepreneurship and Innovation • Honors College • Dean's List**

Study Abroad Experiences:

– **Computer Science at ETH Zürich** (Spring 2022 semester)

*Focus: Human-Computer Interaction (HCI), Mobile Health, 3D Computer Vision*

– Honors College 4-week courses in Italy (May 2019)

## Research Experience

**Graduate Research Assistant**, University of Oxford

*October 2022 – Present*

- Defining psychological safety of autonomous vehicles and developing new interfaces to improve this safety.
- RobotCycle: Collecting and analysing eye gaze data from cyclists to measure the safety of Oxford roads.
- Investigating the experiences of teleoperators of autonomous vehicles.

**Networks Research Project**, Purdue University

*January 2021 – October 2022*

- DARPA Software Defined Networking (SDN) project under supervision of Distinguished Professor Douglas Comer to change the interpretation of MAC addresses to provide reliable VM migration.
- Designed tests utilizing Mininet and ONOS to simulate the architecture and test network performance, working 10 hrs/week under the supervision of Distinguished Professor Douglas Comer.
- Collaborated with a researcher from the Open Networking Foundation to bring industry advice to our work.

## Professional Engineering Experience

**Apple**, San Diego, CA

*January 2025 – July 2025*

*Research and Development Intern – Wireless Technologies and Ecosystems*

**Microsoft**, Redmond, WA

*June 2022 – September 2022*

*Software Engineer Intern – Azure App Service*

- Developed Azure WebJobs for Windows Containers that enable developers to run extra functions on their container sites. Built with C#.

**Microsoft, Redmond, WA**

*May 2021 – August 2021*

*Software Engineer Intern – Azure App Service*

- Designed and implemented Remote Debugging on Azure Windows containers to allow customers to debug NT services and reduce developer costs. Built with C#.
- Future-proofed Remote Debugging to support Visual Studio 2017, 2019, and 2022.
- Architected Azure Site Extensions for Windows containers to deliver greater extensibility and metrics to developers about their containers.

**General Motors, Warren, MI**

*June 2020 – August 2020*

*Vehicle Architecture Intern – Automated Driving*

- Consolidated vehicle sensor alignment with Python, reducing the alignment process from 2-3 weeks to 3-5 days.
- Implemented multicore computation in sensor alignment, resulting in runtimes 2-3 times faster.
- Automated the OS updates for data collection vehicles to decrease the time to test new features.

**General Electric Aviation, Grand Rapids, MI**

*June 2019 – August 2019*

*Software Engineer Intern – Model-Based Development Team*

- Designed algorithms utilizing Python and MySQL to manage Enterprise Architect models.
- Coordinated with Aviage Systems to fulfill Comac C919 aircraft requirements and documentation.

## Teaching & Mentoring

Computer Vision Tutor – Oxford Dept. of Computer Science

*Fall 2025*

Machine Learning Tutor – Oxford Dept. of Computer Science

*Fall 2025*

Law and Computer Science Teaching Assistant – Oxford Dept. of Computer Science

*Fall 2024*

Requirements Engineering Teaching Assistant – Oxford Dept. of Computer Science

*Fall 2023*

Software Group Design Mentor – Oxford Dept. of Computer Science

*Spring 2023, 2024*

C++ Programming Lab Demonstrator – Oxford Dept. of Engineering Science

*Spring 2023, 2024*

Systems Programming Tutor – Purdue CS252, Dept. of Computer Science

*August 2021 – December 2022*

## Merit-Based Awards

- **Capgemini Studentship**

*October 2022 – May 2026*

*Full-tuition scholarship for PhD studies*

- **Lilly Endowment Scholarship**

*August 2018 – May 2022*

Full-tuition scholarship for undergraduate studies awarded to one student in my county for dedication to academics, leadership, community, and extracurriculars.

- **Purdue College of Science Zimmerly Scholarship**

*May 2021*

- **Salesforce Scholarship**

*April 2021*

- **Purdue CS Corporate Partners Scholarship** (Twice Awarded)

*April 2019, April 2020*

## Publications

- (Pre-Print Book Chapter) **Assuring the Trustworthiness of Autonomous Robotic Systems in Challenging Environments**

*Chris Harper, Benjamin Hardin, Lars Kunze*

*Comprehensive Robotics for Extreme and Challenging Environments, Elsevier*

- (Pre-Print) **The Agent Is Not Your Driver: Clarifying the Implied Capabilities of In-Vehicle Anthropomorphic Agents**

*Benjamin Hardin*

- **Engineering Psychological Safety in Autonomous Vehicles: A Systems-Theoretic Framework for Psychological Safety in Autonomous Vehicles and its Validation in Real-World Scenarios**  
Yandika Sirgabsou, **Benjamin Hardin**, François Leblanc, Efi Raili, Pericle Salvini, David Jackson, Marina Jirotko, Lars Kunze  
[PDF](#)
- **What Can Eye Gaze Teach Us About Real-World Cycling? Insights From the Oxford RobotCycle Project**  
**Benjamin Hardin**, Efimia Panagiotaki, Daniele De Martini, Lars Kunze  
[PDF](#)
- **A Survey of Road Scenarios and Their Effects on Autonomous Vehicle Psychological Safety**  
**Benjamin Hardin**, Lars Kunze, Keri Grieman, Marina Jirotko  
[PDF](#)
- **AV-PsySafe: A risk model and analysis method for the psychological safety of human and autonomous vehicles interaction**  
Yandika Sirgabsou, **Benjamin Hardin**, François Leblanc, Efi Raili, Pericle Salvini, David Jackson, Marina Jirotko, Lars Kunze  
Transportation Research Interdisciplinary Perspectives 34  
[PDF](#)
- **The Oxford RobotCycle Project: A Multimodal Urban Cycling Dataset for Assessing the Safety of Vulnerable Road Users**  
Efimia Panagiotaki, Divya Thuremella, Jumana Baghabrah, Samuel Sze, Lanke Frank Tarimo Fu, **Benjamin Hardin**, Tyler Reinmund, Tobit Flatscher, Daniel Marques, Chris Prahacs, Lars Kunze, Daniele De Martini  
IROS 2025, IEEE Transactions on Field Robotics  
[PDF](#)
- (Pre-Print) **A Revised Framework for the Assessment of Psychological Safety in Autonomous Vehicles**  
Yandika Sirgabsou, **Benjamin Hardin**, François Leblanc, Efi Raili, David Jackson, Pericle Salvini, Lars Kunze, Marina Jirotko  
[PDF](#)
- **How well do drivers adapt to remote operation? Learning from remote drivers with on-road experience**  
**Benjamin Hardin**, Pericle Salvini, Marina Jirotko, Lars Kunze  
2024 IEEE Intelligent Vehicles Symposium (IV)  
[PDF](#)
- **Human Involvement in Autonomous Decision-Making Systems. Lessons learned from three case studies in aviation, social care and road vehicles**  
Pericle Salvini, Tyler Reinmund, **Benjamin Hardin**, Keri Grieman, Carolyn Ten Holter, Aaron Johnson, Lars Kunze, Alan Winfield, and Marina Jirotko  
Frontiers in Political Science  
[PDF](#)
- **Safety Assurance Challenges for Autonomous Drones in Underground Mining Environments**  
Philippa Ryan, Arjun Badyal, Samuel Sze, **Benjamin Hardin**, Hasan Bin Firoz, Paulina Lewinska, Victoria Hodge  
Annual Conference Towards Autonomous Robotic Systems  
[PDF](#)
- **A Cross-Sectional Examination of Evictions in Lafayette, Indiana in August 2020 during a Gap in Federal and State Eviction Moreatoria**  
Mary Lang, Justin MacNeill, Ethan Edwards, Ethan Glaser, **Benjamin Hardin**, Jared Huber, Elizabeth Spyrtatos, Jason Ware  
Local Development & Society Journal  
[PDF](#)
- **DCnet: Evaluation of a New Data Center Architecture**  
**Benjamin Hardin**, Douglas Comer, Adib Rastegarnia  
2023 Innovation in Clouds, Internet, and Networks (ICIN)  
[PDF](#)

- **On the Unreliability of Network Simulation Results From Mininet and iPerf**

*Benjamin Hardin, Douglas Comer, Adib Rastegarnia*

2023 International Conference on Computer, Control, and Robotics (ICCCR)

[PDF](#)

## Community Engagement

### **Development Officer – Wolfson College Boat Club, Oxford**

*June 2026 – Present*

- Responsible for alumni engagement and outreach.

### **Mentor – RISE Global Program**

*Fall 2024 – Present*

- I mentor high school students from around the world guiding topics such as choosing education and careers, finding passions, and goal setting.

### **Tutor & Supervisor – Oxford Prospects Programme**

*Fall 2023 – Present*

- I work with Chinese exchange students to review statements of purpose and graduate applications, guide development of their projects during their exchange visit, and share my perspectives on life in Oxford and the application process.

### **President – Wolfson College Boat Club, Oxford**

*June 2024 – June 2025*

- Oversaw a 17-person committee managing the boat club, welfare, its finances, training, and outreach.

### **Men's Captain – Wolfson College Boat Club, Oxford**

*June 2023 – June 2024*

- Led scheduling of training for over 80 men's club members
- Oversaw hiring, management, and payment of coaches
- Mentored and advised club members

### **President – Purdue Honors College App Development Committee**

*August 2018 – June 2022*

- Guided teams to develop and maintain a client-server app to over 900 students and faculty to encourage community involvement by incentivising academic achievements, event attendance, and volunteer work.
- Grew development team from 4 members to 20 members and guided 34 app update releases.
- Developed Firebase API endpoint, creating a single request framework for iOS, Android, and web apps.
- Personally built and integrated in-app messaging and a laundry machine status based on user feedback.

### **President – Lilly Scholars Network Purdue**

*August 2018 – January 2022*

- Led volunteering events in the community and statewide with Lilly Scholars from Indiana.
- Held monthly professional events to connect members and foster a scholar community at Purdue.

### **Marketing & Design Executive – Purdue Association of IT Professionals**

*September 2018 – May 2019*

- Created over 30 designs for shirts, flyers, and graphics to promote club spirit and outreach.
- Mentored members to foster professional development.
- Worked 7+ hrs/week to plan events and create materials to benefit the club's impact and future.

### **Eagle Scout – Boy Scouts of America**

*June 2012 – June 2019*

- 200 man-hour landscaping project for the North Madison Church of Christ in Madison, IN.
- Raised funds, drafted and revised plans, selected plants, and organised labor to accomplish the project.

### **Outdoor Classroom Designer – Historic Eleutherian College, Inc.**

*July 2021 – August 2021*

- Sought out a Lilly Scholars Network Lead Forward Grant, searched for community needs by talking to high school teaches, proposed the project, and was awarded \$2500.
- Designed, built, and installed an outdoor classroom for 24 students to facilitate STEM training for minority student teachers.
- Over 155 man-hours in total.